

Name of Student:

Scholar No. :

Indian Institute of Information Technology, Bhopal

Mid Term Examination, Sep. 2025

Course: BTech(IT)

Semester: III

Subject: Design Analysis & Algorithm

Subject Code: IT-3004

Time: 90 Minutes

Max. Marks: 20

Note: Attempt all questions. All questions carry equal marks.

Q. No.	Questions	Marks
1.	i) What is the time complexity of the below function? <pre>void fun(int n, int arr[]) { int i = 0, j = 0; for (; i < n; ++i) while (j < n && arr[i] < arr[j]) j++; }</pre> ii) Calculate the time complexity of this section of code. Assume that n is divisible by <pre>j=1; while(j<=n/2){ i=1; while(i<= j){ printf("%d%d", i,j); i++; } j++; }</pre>	4
2.	i) Consider the following recurrence relation: $T(n) = \sqrt{n}T(\sqrt{n}) + n$ for $n \geq 1$, 1 for $n = 1$. Determine the time complexity and provide a justification for your answer. ii) Use a recursion tree method to give an asymptotic tight solution to the recurrence $T(n) = T(n/2) + T(n/4) + T(n/8) + n$	4
3.	Maximum subarray Sum problem is to find the subarray with maximum sum. For example, given an array {12, -13, -5, 25, -20, 30, 10}, the maximum subarray sum is 45. The naive solution for this problem is to calculate sum of all subarrays starting with every element and return the maximum of all. We can solve this using Divide and Conquer, what will be the worst case time complexity using Divide and Conquer?	4
4.	i) Order the following function by growth rate: N, $\sqrt{N}$, $N^{1.5}$, N^2, $N \log N$, $N \log \log N$, $N \log N^2$, $2/N$, 2^N. Indicate if any functions grow at the same rate. ii) Given a sorted array of integers, what can be the minimum worst-case time complexity to find ceiling of a number x in given array? The ceiling of an element x is the smallest element present in array which is greater than or equal to x. Ceiling is not present if x is greater than the maximum element present in array. For example, if the given array is {12, 67, 90, 100, 300, 399} and $x = 95$, then the output should be 100.	4
5.	Suppose we are sorting an array of eight integers using heapsort, and we have just finished some heapify (either max heapify or min heapify) operations. The array now looks like this: 16 14 15 10 12 27 28 How many heapify operations have been performed on root of heap?	4